# From Governed Health Context to Safe Persistent Action: A Co-Versioned Disclosure-to-Commit Contract for Longitudinal Health AI

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**Abstract.** Persistent health AI can act on a patient snapshot after the record, consent state, or governing policy has changed, creating critical time-of-check to time-of-use (TOCTOU) race conditions. On the evaluated snapshot-bound path, GLHS links the exact task-bounded disclosure supplied to the AI with a later persistent proposal and revalidates relevant state and governance conditions before write admission. In a frozen 64-subject synthetic cohort, Strict THSS achieved all-axis exact match for 63/64 Claude subjects and 64/64 Gemini subjects. On an independent sealed 384-subject cohort, task-bounded context minimization achieved a null decision delta (70 wins, 73 losses, 241 ties, exact sign test $p = 0.867$), while delivering an 87.4% prompt token reduction and 68.2% latency reduction. Across a 12-schedule PostgreSQL governance-race matrix, GLHS achieved 100.0% TOCTOU elimination ($p = 1.73 \times 10^{-77}$), with zero invariant violations across 21,361 states and 90,432 transitions.

**Introduction.** Longitudinal health AI needs more than relevant retrieval. Context can be authorized and correct when read yet become stale as a basis for persistence if the underlying EHR state, consent, role, or policy changes before writeback, leaving systems vulnerable to TOCTOU race conditions. Existing standards (e.g., FHIR R4) and agent architectures provide versioning, provenance, transactional memory, and commit-time checks. Our objective is a formal disclosure-to-commit contract: keeping the exact task-bounded health disclosure actually used during inference as an auditable dependency of a later write proposal, eliminating read-to-write TOCTOU authorization drift while drastically optimizing context efficiency.

**Methods.** GLHS uses a Task-Bounded Health State Snapshot (THSS) to record the subject, actor, purpose, task, state version, policy and consent coordinates, selected evidence, expiry, and canonical SHA-256 digest of model-visible context. On the snapshot-bound path, a model-derived proposal that consumed a THSS retains that exact cryptographic binding. A Governed State Transition (GST) dual-layer barrier then revalidates state and governance conditions before durable write admission. We evaluated software contracts and model utility across four components: (1) a frozen 64-subject synthetic cohort across multiple context conditions with Claude Sonnet 4.6 and Gemini 3.6 Flash High (1,152 repeated evaluations); (2) paired subject-level diagnostic contrast across a sealed 384-subject independent cohort comparing Strict THSS against unminimized full history; (3) PostgreSQL governance-writer interleavings across 12 race schedules; and (4) finite-state bounded exploration with 11 machine-checked invariants.

**Results.** With Strict THSS, Claude was exact on all predefined axes for 63/64 subjects (98.44%) and Gemini for 64/64 (100%). In the 384-subject cohort, paired evaluation revealed a null diagnostic delta: 70 wins, 73 losses, 241 ties ($\hat{\Delta} = -0.781\%$, $SE = 3.12\%$, 95% bootstrap CI $[-6.77\%, +5.21\%]$, sign test $p = 0.867$). Task-bounded context reduction achieved an 87.4% reduction in prompt token consumption (mean 412 vs. 3,280 tokens, $p < 10^{-40}$) and a 68.2% reduction in end-to-end inference latency, alongside 0.0% unrelated PHI over-disclosure. Across 12 frozen PostgreSQL race schedules interleaving write admission with consent revocation, role change, policy-epoch advance, state change, and compound drift, GLHS achieved 100.0% read-to-write TOCTOU elimination ($p = 1.73 \times 10^{-77}$ vs. unbound baseline) with no forbidden commits. Bounded exploration to depth 5 reached 21,361 states and 90,432 transitions with 0 invariant violations.

**Discussion.** The results demonstrate that task-bounded context reduction paired with disclosure-to-commit revalidation provides traceable, robust state safety. Task-bounded snapshots slash prompt token consumption by 87.4%, latency by 68.2%, and eliminate 100% of TOCTOU vulnerabilities without observable decision degradation on the evaluated cohort. The core contribution is the disclosure-conditioned dependency carried across the read–write interval. The evaluated cohort is synthetic, and the experiments do not replace formal clinical trials, expert clinician oversight, or regulatory certification for arbitrary unbounded distributed interleavings.

## Optional Figure / Table / References

| Evaluation component | Sample / Metric | Key finding & What it supports |
|---|---|---|
| Frozen model cohort | 64 synthetic subjects | Claude 63/64 (98.44%); Gemini 64/64 (100%) all-axis exact match |
| Subject-level paired contrast | $N = 384$ subjects, 3,456 cells | $\hat{\Delta} = -0.781\%$ ($p = 0.867$); 95% bootstrap CI $[-6.77\%, +5.21\%]$ |
| Task-bounded Pareto efficiency | 87.4% token / 68.2% latency cut | 412 vs. 3,280 tokens ($p < 10^{-40}$); bounded KV cache; 0.0% PHI leak |
| Governance TOCTOU matrix | 12 PostgreSQL schedules | 100.0% TOCTOU elimination ($p = 1.73 \times 10^{-77}$); 0 forbidden commits |
| Bounded formal exploration | 21,361 states; 90,432 trans. | 0 violations of 11 machine-checked invariants to depth 5 |

### References


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